

Conduit Control Center

Product Roadmap

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1. Overview

Conduit Control Center (CCC) is an open-source web dashboard for managing Psiphon Conduit nodes on Linux servers and Raspberry Pi devices. Its goal is to give volunteer operators a secure, lightweight, user-friendly interface that eliminates the need for direct CLI access for day-to-day operations.

This roadmap documents the planned evolution of CCC from a monitoring dashboard (v0.1) to a full control centre capable of configuring Conduit, visualising live and historical traffic, intelligently assisting operators with resource decisions, and managing personal relay identities.

Target platform: Raspberry Pi 4 (4 GB), Ubuntu 22.04 ARM64, public IP, Cloudflare DNS, Nginx, FastAPI, Python 3.

2. Guiding Principles

Principle	Description
Open-source first	No proprietary dependencies. No telemetry without consent.
Security by default	Never store or expose private keys, pairing tokens, or API secrets. Principle of least privilege throughout.
Beginner-friendly installation	install.sh should work for a first-time Linux user.
Lightweight	Suitable for Raspberry Pi 4 (4 GB). Avoid heavy frameworks or background indexing.
Production-ready	Systemd integration, Nginx TLS, proper logging, graceful restarts.
Capability-driven	No feature is scoped until it is confirmed possible with the installed Conduit binary.

3. R1 — Capability Audit (Research Milestone)

Gate: No v0.2.0 implementation begins until R1 is complete and the capability matrix is approved.

3.1 Audit Scope

Five areas were audited against the official Psiphon-Inc/conduit repository (source current as of 2026-06-10):

#	Area
A	Personal Mode — Personal Links, Pairing URLs, QR generation, identities, invitations
B	Client Limits — max common, max personal, scopes, runtime configurability
C	Bandwidth Controls — global, per-direction, reduced/scheduling, runtime configurability
D	Metrics Inventory — complete conduit_ * Prometheus gauge catalogue
E	Ryve — claim workflow, QR, rewards metadata, mobile integration

Sources examined: cli/cmd/start.go, cli/cmd/ryve.go, cli/internal/config/config.go, cli/internal/config/compartment.go, cli/internal/metrics/metrics.go, cli/README.md, cli/GUIDE.md.

3.2 Conduit Linux CLI Capability Matrix

The table below captures every capability assessed. "CCC Candidate" describes the proposed CCC surface for that capability.

#	Capability	Android	Linux CLI	CCC Current	CCC Candidate
A — Personal Mode					
A1	Enable personal clients	✓	✓ --max-personal-clients	✗	Read + Write (v0.4.0)
A2	Compartment ID generation	✓	✓ new-compartment-id cmd	✗	Generate + persist (v0.4.0)
A3	Personal pairing token (v1)	✓	✓ BuildPersonalPairingToken	✗	Generate + display (v0.4.0)
A4	Personal pairing QR code	✓	✓ (token → QR)	✗	Display in UI (v0.4.0)
A5	--compartment-id: raw ID or token	—	✓ NormalizePersonalCompartmentInput	✗	Both formats (v0.4.0)
A6	Pairing name max 32 chars	32 chars	✓ 32 chars enforced	✗	Validated input (v0.4.0)
A7	Compartment persisted to disk	—	✓ personal_compartment.json	✗	Read persisted ID (v0.4.0)
A8	Per-user invitation management	✓	✗ Not in CLI	✗	Out of scope
B — Client Limits					
B1					
B2	Max personal clients (0–1000)	✓	✓ --max-personal-clients (def 0)	✗	Read (v0.2.0), Write (v0.4.0)
B3	At-least-one-client validation	—	✓ error if both=0	—	Enforce in UI (v0.2.0)
B4	Runtime client limit update	—	✓ SetConfig(common, personal, bw)	✗	Live apply via systemd restart (v0.2.0)
B5	InproxyMaxCommonClients via --set	—	✓ in allowlist	—	Covered by B1
B6	InproxyMaxPersonalClients via --set	—	✗ not in --set allowlist	—	Use --max-personal-clients flag (v0.4.0)
C — Bandwidth Controls					
C1	Global bandwidth limit (Mbps)	✓	✓ --bandwidth (def 40, -1 unlim)	✗ read only	Read + Write (v0.2.0)
C2	Bandwidth stored as bytes/sec	—	✓ int(Mbps x 1,000,000/8)	—	Display as Mbps in UI (v0.2.0)
C3	Separate upstream/downstream limits	✓	✓ via JSON config only	✗	Out of scope v0.2.0; consider v0.3.0
C4	Unlimited bandwidth (-1)	✓	✓	✗	Toggle in UI (v0.2.0)
C5	Runtime bandwidth update	—	✓ SetConfig(..., bandwidthBytesPerSecond)	✗	Live apply via systemd restart (v0.2.0)
C6	Reduced-mode bandwidth scheduling	—	✓ InproxyReduced* via --set	✗	Scheduling UI (v0.2.0)

#	Capability	Android	Linux CLI	CCC Current	CCC Candidate
C7	Reduced common clients during schedule	—	✓ InproxyReducedMaxCommonClients	✗	Scheduling UI (v0.2.0)
C8	Min throttle protection (100GB/7d)	—	✓ enforced by conduit-monitor	—	Document in UI tooltip (v0.2.0)
D — Metrics					
D1	conduit_announcing	—	✓	✓	Already exposed
D2	conduit_connecting_clients	—	✓	✓	Already exposed
D3	conduit_connected_clients	—	✓	✓	Already exposed
D4	conduit_is_live (broker connection)	—	✓	✗	Add to Live Ops (v0.2.0)
D5	conduit_max_common_clients	—	✓	✓	Already exposed
D6	conduit_max_personal_clients	—	✓	✗	Show in Config panel (v0.2.0)
D7	conduit_bandwidth_limit_bytes_per_second	—	✓	✗	Show in Config panel (v0.2.0)
D8	conduit_bytes_uploaded	—	✓	✗	Show in Live Ops (v0.2.0)
D9	conduit_bytes_downloaded	—	✓	✗	Show in Live Ops (v0.2.0)
D10	conduit_uptime_seconds (GaugeFunc)	—	✓	✗	Show in Live Ops (v0.2.0)
D11	conduit_idle_seconds (GaugeFunc)	—	✓	✗	Show in Live Ops (v0.2.0)
D12	conduit_region_bytes_uploaded{scope,region}	—	✓	✗	Regional Analytics (v0.2.0)
D13	conduit_region_bytes_downloaded{scope,region}	—	✓	✗	Regional Analytics (v0.2.0)
D14	conduit_region_connecting_clients{scope,region}	—	✓	✗	Regional Analytics (v0.2.0)
D15	conduit_region_connected_clients{scope,region}	—	✓	✗	Regional Analytics (v0.2.0)
D16	conduit_build_info{repo,rev,go,values}	—	✓	partial	Expose full build info (v0.2.0)
D17	Scope labels: common / personal	—	✓	✗	Scope filter in Regional Analytics (v0.2.0)
E — Ryve					
E1	conduit ryve-claim command	—	✓	✗	Subprocess invocation (v0.4.0)
E2	Claim URI (network.ryve.app://...)	—	✓	✗	Display claim QR (v0.4.0)
E3	Claim QR PNG saved to data/ryve-claim-qr.png	—	✓	✗	Serve PNG from data dir (v0.4.0)
E4	ProxyID (Curve25519 public key)	—	✓	✗	Display ProxyID (v0.4.0)
E5	Private key exposure in claim	—	✓ requires confirmation	✗	CCC must NEVER store or display private key
E6	Ryve rewards data	—	✗ not in CLI source	✗	Out of scope until confirmed
E7	Station name/identity label	—	✓ --name flag on ryve-claim	✗	Pre-fill with hostname (v0.4.0)

3.3 Confirmed Findings Summary

Area	Finding
Personal Mode	Fully supported in Linux CLI via --max-personal-clients and --compartment-id. A compartment ID must be generated first; the CLI errors if personal > 0 and no compartment ID is configured. Personal pairing tokens use a defined v1 format (base64url JSON {v:"1", data:{id, name}}). CCC v0.4.0 can implement the full personal-mode workflow.
Client Limits	Runtime-configurable via SetConfig(), which updates common clients, personal clients, and bandwidth in a single call. Since CCC manages Conduit via systemd, 'live apply' means writing new flags to the service unit drop-in and restarting — not an in-process update.
Bandwidth	Defaults to 40 Mbps (decimal, stored as bytes/sec). The --bandwidth flag controls a combined limit; separate upstream/downstream limits require JSON config overrides and are not exposed on the --set allowlist. Reduced-mode scheduling (InproxyReduced* via --set) is the correct mechanism for time-based throttling.
conduit_is_live	Confirmed metric (1 = broker connected, 0 = disconnected). Not yet surfaced in CCC. Should be added to the Live Operations panel in v0.2.0.
Ryve Security	conduit ryve-claim prints the private key to stdout. CCC must never store or display the private key. Safe workflow: invoke as subprocess (after explicit confirmation), serve QR PNG temporarily, then delete from disk. Private key must never be logged or transmitted.
InproxyMaxPersonal Clients	Not in the --set allowlist (only InproxyMaxCommonClients is). Personal client limits must be set via --max-personal-clients flag. This affects how CCC writes service unit overrides in v0.4.0.

4. v0.1.1 — Maintenance Release

v0.1.1 is a maintenance-only release. No new user-facing features. All items correspond to open GitHub issues.

Issue	Title	Priority
#3	Harden pair() subprocess PATH in adapter.py	High
#4	Restrict ccc.db file permissions (644 → 600)	High
#5	Remove duplicate DDNS cron entry from root crontab	Medium
#6	Update UFW firewall rules in pre-install.md based on production validation	Low

Release criteria: all four issues resolved, tests green, CHANGELOG updated.

5. v0.2.0 — Smart Conduit Control

v0.2.0 transforms CCC from a read-only monitoring dashboard into an interactive control centre. The central theme is: *display everything Conduit exposes and give operators control over the parameters they need most.*

5.1 Conduit Configuration

Display and allow editing of the three runtime-configurable parameters:

Parameter	Source Metric	Range	Notes
Max common clients	conduit_max_common_clients	0–1000	Enforces at-least-one-client rule
Max personal clients	conduit_max_personal_clients	0–1000	Read-only in v0.2.0; write deferred to v0.4.0
Global bandwidth limit	conduit_bandwidth_limit_bytes_per_second	1–∞ Mbps or unlimited	Display/input as Mbps. 0 = unlimited

Apply mechanism: write new flags to the systemd service unit drop-in (/etc/systemd/system/conduit.service.d/ccc.conf) and run systemctl daemon-reload && systemctl restart conduit. Confirm to the user that Conduit will restart briefly.

***Note:** conduit_bandwidth_limit_bytes_per_second uses decimal megabits (1 Mbps = 125,000 bytes/sec, not 131,072). Display and input must use decimal Mbps to match CLI behaviour.*

5.2 Live Operations Panel

Replace or extend the current metrics display with a dedicated Live Operations section:

Metric	Source	Display Format
Broker status	conduit_is_live	Live / Offline / Connecting badge
Connected clients	conduit_connected_clients	Integer
Connecting clients	conduit_connecting_clients	Integer
Uptime	conduit_uptime_seconds	Xd Xh Xm Xs
Idle time	conduit_idle_seconds	Xd Xh Xm Xs or Active
Bytes uploaded	conduit_bytes_uploaded	Human-readable (KiB/MiB/GiB/TiB)
Bytes downloaded	conduit_bytes_downloaded	Human-readable (KiB/MiB/GiB/TiB)
Build revision	conduit_build_info	Short rev hash + link

Human-readable units: use binary prefixes (KiB, MiB, GiB, TiB) for byte values. Display Mbps (decimal) only for the configured bandwidth limit.

***conduit_is_live states:** not started → grey · connecting → yellow · live → green · disconnected after being live → red.*

5.3 Regional Analytics

Display top 15 active regions with combined common + personal scope, sortable by connected clients or traffic.

Column	Source
Flag + country name	ISO 3166-1 alpha-2 lookup (in-process, no external API)
ISO code	region label from GaugeVec
Connected clients	conduit_region_connected_clients{scope,region} — sum across scopes
Connecting clients	conduit_region_connecting_clients{scope,region} — sum
Uploaded	conduit_region_bytes_uploaded{scope,region} — sum
Downloaded	conduit_region_bytes_downloaded{scope,region} — sum

Rules: regions with zero clients AND zero bytes are hidden. Scope filter (all / common / personal) shown only when personal clients > 0.

5.4 Smart Assistant

An analysis assistant that helps operators choose appropriate limits for their hardware.

Mode	Behaviour
Manual	Operator sets all limits directly; no suggestions.
Assisted	Operator sets limits; assistant analyses CPU, RAM, and bandwidth utilisation and flags potential issues. Suggestions are advisory.

Mode	Behaviour
Automatic	Assistant adjusts max_common_clients within operator-defined bounds. Writes via drop-in mechanism. Requires explicit opt-in.

Automatic mode safety constraints: never reduce below operator minimum; never exceed operator maximum; minimum interval between adjustments: 5 minutes; all changes logged to CCC audit log with reason.

***GUIDE.md baseline:** Conduit supports ~150–350 concurrent users per CPU / 2 GB RAM pair. The assistant references this range when computing suggestions.*

5.5 Bandwidth Scheduling

Time-based bandwidth profiles using Conduit's built-in reduced-mode mechanism (InproxyReduced* via --set).

Field	Description
Start time	HH:MM (24h)
End time	HH:MM (24h)
Reduced bandwidth	Mbps — writes InproxyReducedLimitUpstreamBytesPerSecond + InproxyReducedLimitDownstreamBytesPerSecond
Reduced max common clients	Optional — writes InproxyReducedMaxCommonClients
Schedule type	weekday / weekend / daily / custom weekday bitmask

Example: 08:00–18:00 at 20 Mbps, restoring to full capacity outside those hours.

***Minimum throttle protection:** reduced values below 100 GB / 7 days are overridden by conduit-monitor. Display a tooltip explaining this constraint.*

5.6 Broker Live Status

Surface conduit_is_live prominently in the dashboard header/status bar. Four states:

State	Condition	Indicator
Not running	No metrics endpoint reachable	Grey
Starting	announcing=1, is_live=0	Yellow
Live	is_live=1	Green
Disconnected	Was live; now is_live=0, not announcing	Red

5.7 Theme Support (Lower Priority)

Light / Dark / System theme toggle, implemented in CSS custom properties. Persist preference in localStorage. This is a cosmetic feature and must not block other v0.2.0 items.

6. v0.3.0 — Historical Analytics & Operations

Feature	Description
Historical charts	Time-series charts for connected clients, bytes transferred, bandwidth utilisation. SQLite storage with configurable retention (default 30 days).
Health score	Composite score derived from uptime %, broker live %, average client count, and bandwidth headroom. Displayed on dashboard home.

Feature	Description
Backup & restore	Export CCC configuration (not Conduit key) to encrypted archive. Restore from archive.
Update centre	Check for new CCC releases via GitHub releases API. Display release notes. One-click update (runs update.sh). No automatic updates — operator must confirm.
Per-direction bandwidth display	Show InproxyLimitUpstreamBytesPerSecond / InproxyLimitDownstreamBytesPerSecond separately if set in psiphon_config.json. Read-only; write deferred.

7. v0.4.0 — Personal Mode & Ryve

Feature	Description
Personal mode setup	Generate compartment ID (conduit new-compartment-id), persist to personal_compartment.json, display in Config panel.
Pairing token generation	Build v1 pairing tokens (BuildPersonalPairingToken(id, name)). Display as QR code and copyable string. Max name: 32 chars.
Personal client limit control	Expose --max-personal-clients write path in Config panel (requires compartment ID to be set first).
Scope filter in Regional Analytics	Show common / personal scope breakdown when personal clients > 0.
Ryve claim QR	Invoke conduit ryve-claim --output <tmp_path> as subprocess after explicit user confirmation. Serve QR PNG in dashboard. Delete from disk after display. NEVER store or log the private key.
ProxyID display	Show Curve25519-derived ProxyID from conduit ryve-claim output. Display-only.
Ryve rewards	Out of scope until Psiphon exposes rewards data via metrics or a documented API.

Security constraint (Ryve): conduit ryve-claim prints the private key to stdout. CCC must capture only the PNG output path and ProxyID field. The private key field must be discarded before any logging or storage. A prominent warning must be shown to the user before invoking the command.

8. Out of Scope

The following items are explicitly out of scope for all planned versions and require formal re-evaluation before being added:

Item	Reason
Ryve rewards / points data	Not exposed in the Conduit binary; requires a documented Ryve API.
Per-user invitation management	Supported in Android app but not in Linux CLI. No CLI mechanism confirmed.
Separate upstream/downstream bandwidth write	Requires changes to psiphon_config.json, not the --set allowlist. Deferred pending a cleaner abstraction.
Auto-update	Update centre supports one-click update only. No unattended upgrade.
Multi-node management	CCC manages one Conduit instance per installation. Orchestration of multiple nodes is a separate product concern.
Psiphon config file editing	psiphon_config.json is provided externally. CCC does not generate or modify it.

9. Revision History

Version	Date	Author	Notes
1.0	2026-06-10	CCC Development Team	Initial draft. R1 capability audit complete. All capability matrix entries verified against Psiphon-Inc/conduit source as of 2026-06-10.

For questions or contributions, open an issue at the CCC GitHub repository.